Near-Degeneracy-Aware OLA Inversions Applied to *Kepler* Red-Giant Observations

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1. Context

Angular momentum transport (AMT) inside stars remains one of the open problems of stellar physics [1]. The mechanisms redistributing angular momentum as stars evolve are still poorly constrained. Asteroseismology offers a unique observational probe on this question as it allows us to directly measure **internal rotation** (e.g. [2]).

2. Asteroseismology

→ Probes stellar interiors and dynamics through resonant **oscillations.**

→ Two mode types: pressure **(p) modes**, sensitive to the outer envelope, and gravity **(g) modes**, confined to the deep radiative interior.

→ **Red giants** are prime targets: their **mixed p–g modes** [2] give direct access to the deep core, unlike other stars where g-modes stay evanescent in the envelope.

3. Near degeneracies

Standard **rotational inversions** assume a linear, symmetric splitting of mixed modes by rotation (e.g. [3]). This assumption breaks down when the rotation rate approaches the frequency spacing between **neighbouring modes**: the mode becomes near-degenerate, inducing an **asymmetry** (e.g. [4,5,6,7], posters 72 & 89, Fig. 1). The distance to the central peak is no longer equal for **splittings of different m**, and can even involve overlap between peaks of different modes.

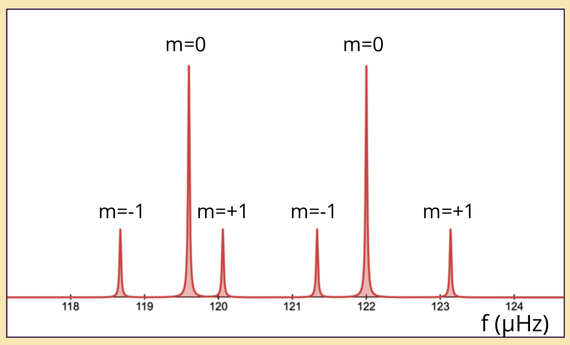


Fig. 1: Evidence of near-degeneracy effects on two ℓ = 1 multiplets

Neglecting these effects in the inversion of the rotation profile introduces a **bias**, which increases with **stellar evolution and rotation rate**. [6]

4. Inversion Pipeline

Quadratic eigenvalue problem (QEP, e.g. [6,7]):

$$\omega^2 \mathbb{I}\vec{a} - \omega \mathbf{R}\vec{a} - \mathbf{L}\vec{a} = 0$$

Exact solution of the **QEP** with adiabatic tracking of the eigenvalues

Kernel computation

See details and tests on **synthetic data** in poster P-072 and [8]

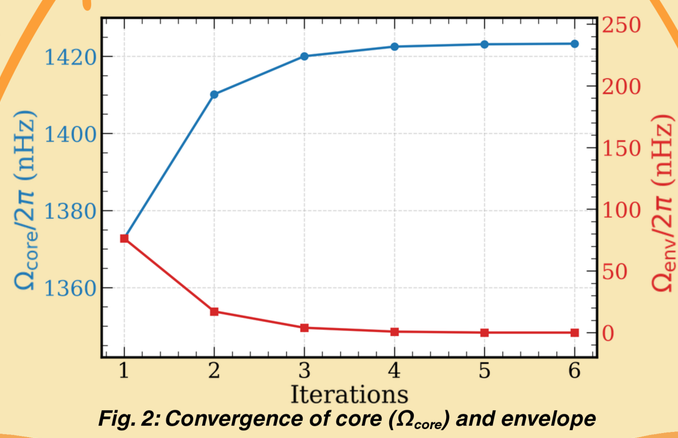


Fig. 2: Convergence of core (Ω_{core}) and envelope (Ω_{env}) rotation rates over inversion iterations.

Solving the QEP

eMOLA* [9]

*extended Multiplicative Optimally Localized Average

5. Target and Model

Based on the catalog of [10], we selected a high signal-to-noise target with a **rapidly rotating core**, such that the core rotation rate reported is comparable to the typical frequency spacing between mixed modes. We **fitted** the oscillation modes using the apollinaire package [11], with initial frequency guesses and mode identification provided by the BACH asymptotic code [12] (Fig. 3).

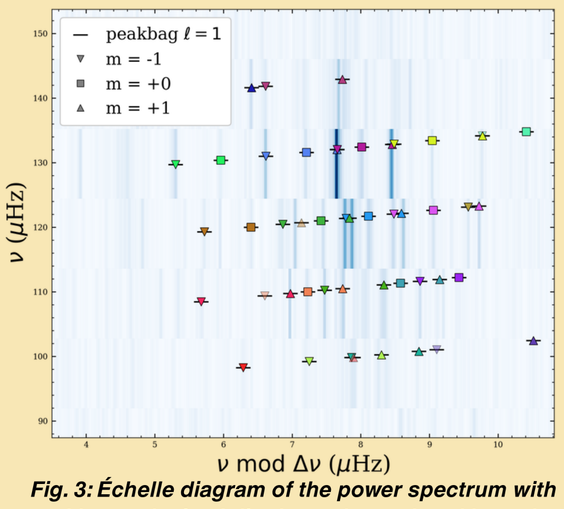


Fig. 3: Échelle diagram of the power spectrum with peakbagged mixed dipole modes, colored by order.

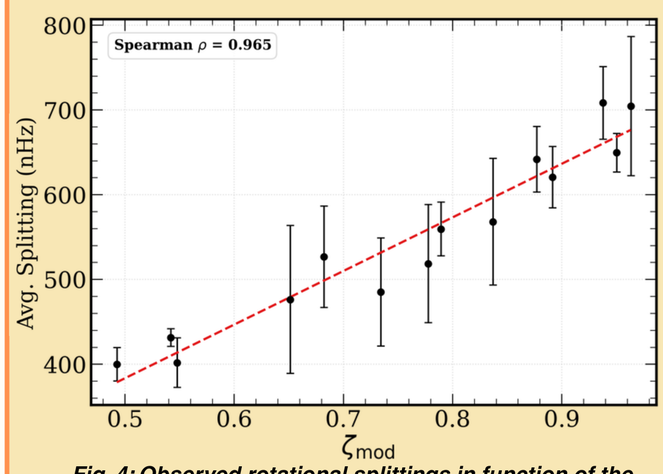


Fig. 4: Observed rotational splittings in function of the mode trapping parameter ζ

We used the simplex search in the MESA test suite (with GYRE for mode determination) [13,14,15] to find the **model minimizing χ^2** from spectroscopic and asteroseismic constraints, and verified a **strong correlation** between model ζ (core-to-total inertia ratio) and observed **average rotational splittings** (Fig. 4), a key requirement for reliable rotational inversions [16].

6. Results/Inversion

- The inclusion of near-degeneracies had a **non negligible impact** on our inversion.
- We observed **rapid convergence** of the iterative eMOLA method even in the absence of good initial guess and with **real data.**
- This real case study **confirms the bias predictions** made by [6].
- We have coded an **automatic pipeline** going from average splitting and model to near-degeneracy aware inversion in one click.

method	Ω_{core} (nHz)	Ω_{env} (nHz)
Classical	**1372 ± 43**	**76 ± 42**
ND-aware	**1423 ± 46**	**0 ± 52**

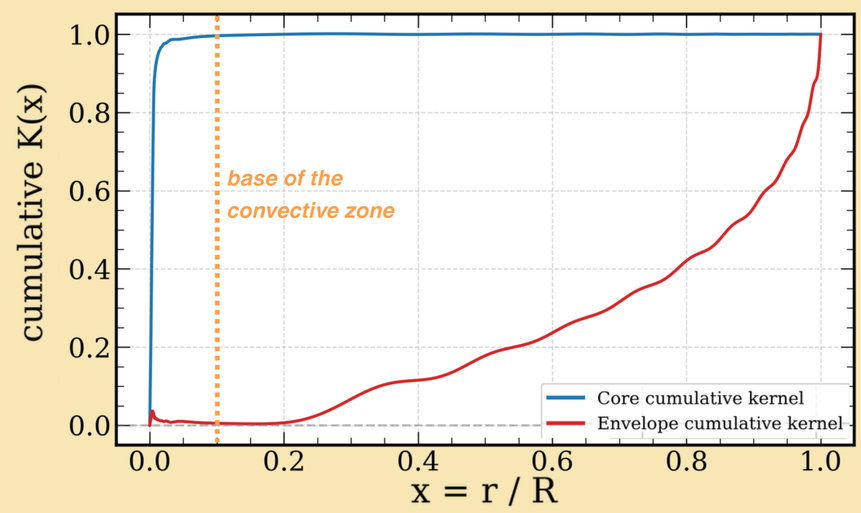


Fig. 5: Cumulative rotation kernels for the target star

Perspectives

- Apply this method to a **bigger sample** of stars with a custom grid of models.
- Write the pipeline in a compiled language to enhance speed.

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[1]Aerts+ 2019, ARA [2]Beck+ 2012, Nature [3]Gough & Thompson 1990 [4]Dziembowski+ 1992, ApJ [5]Deheuvels+ 2017, A&A [6]Ahlborn+ 2025b, A&A [7]Liagre+ 2026, A&A [8] Ahlborn [...] **Perrin**+ in prep [9] Ahlborn+2022, A&A [10]Gehan+ 2018, A&A [11]Breton+ 2022, A&A[12] Liagre+ priv. comm. [13]Paxton+ 2011-2019, ApJS [14]Jermyn+ 2023, ApJS [15]Townsend+ 2013,2018, MNRAS [16]Ahlborn+ 2025, A&A

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